

Sleep Health and Lifestyle Pattern Discovery

M4 Final Report | CS 4412 Data Mining
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Executive Summary. In the current study, a real-life sleep health dataset was used to find health-lifestyle clusters, the characteristics that differentiate between them, and any outliers. Both K-Means and Agglomerative Clustering algorithms found six clusters (Silhouette score is 0.537 and 0.540, respectively). PCA maintained 69.9% of variance on two dimensions, while LOF detected 18 outliers. The most important differentiating variables included systolic blood pressure, heart rate, stress, and physical activity. This analysis shows that sleep health patterns in the chosen dataset can be best described by a more comprehensive approach to health status.

1. Introduction

The final report provides the finalized version of the data mining assignment aimed at discovering patterns in the sleep health and lifestyle data. Although initially, an artificial dataset simulating fatigue management based on two conditions - power naps and coffee - was chosen for the assignment, after getting feedback from peers regarding the implementation of M2, the artificial nature of the data was recognized as being a drawback since it made the results less realistic. Thus, for the final data set, a more natural Sleep Health and Lifestyle database was chosen to ensure the authenticity of the discovery story.

The number of records in the final dataset is equal to 374. Such attributes as duration of sleep, sleep quality, physical activity level, stress level, heart rate, number of steps per day, body mass index category, occupation, presence of sleep disorder, and blood pressure were included. Blood pressure was decomposed into two variables – systolic and diastolic.

Discovery Questions

- DQ1: What natural sleep-health segments exist in the data?
- DQ2: Which health and lifestyle features distinguish healthier or less healthy profiles?
- DQ3: Which profiles are unusual compared with the rest of the population?

However, the purpose of this project was neither to create a tool for diagnosing patients nor predict using the models. Rather, the objective was one of discovery where the objective was to find out interpretive patterns and their significance, in relation to the domain.

2. Methods

This analysis adhered to a typical pattern for data mining tasks that includes data pre-processing, data preparation, exploration, clustering, visualization, interpretation, outlier detection, and critical evaluation. The main pre-processing steps involved removal of duplicate records, replacement of missing records in "sleep disorders" field with value "None," normalization of numeric attributes for similarity measure purposes, and segregation of categorical attributes for interpretation.

Technique	Parameter / Role	Reason for Use
K-Means	k = 2-6 tested; final k = 6	Primary centroid-based segmentation method with interpretable profiles.
Agglomerative Clustering	n_clusters = 6	Cross-method validation using hierarchical structure rather than centroids.
DBSCAN	eps = 1.2, min_samples = 5	Density-based contrast; tested whether irregular dense regions or noise dominated.
PCA	2 components	Visualization of cluster separation in lower-dimensional space.
Decision Tree / Naive Bayes	Decision tree max_depth = 4	Interpretability tools for explaining K-Means labels, not independent validation.
Local Outlier Factor	n_neighbors = 20, contamination = 0.05	Detection of profiles unusual relative to local neighborhoods.

3. Results: Patterns Discovered

3.1 Clustering Structure

Silhouette score was used to test the performance of the K-Means model from $k = 2$ to $k = 6$. As the value of k increases, there was an increasing performance as indicated by an increase in silhouette scores. The highest silhouette score for the K-Means technique was observed when $k = 6$ with a score of 0.537. A six-cluster Agglomerative Clustering gave a silhouette score of 0.540, which was almost identical to the K-Means technique. The significance of this is that the two models have completely different ways of operation: K-Means uses centroids while Agglomerative builds a hierarchy through mergers.

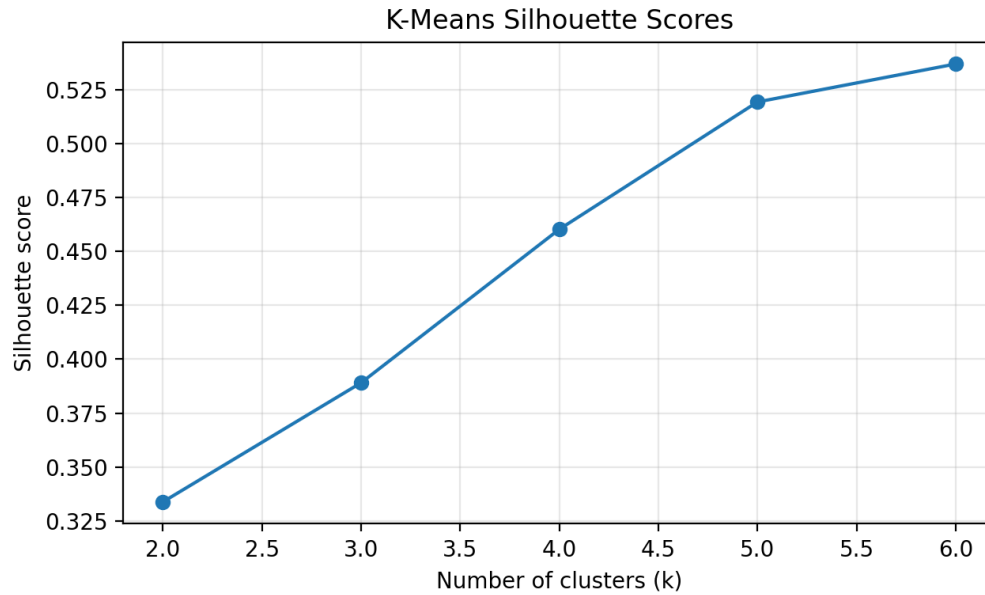


Figure 1. K-Means silhouette scores increased through $k = 6$, supporting a six-cluster solution.

Cluster	Segment Name	Sleep	Quality	Stress	Activity	Heart Rate	Sys. BP
0	Balanced, Active, Lower-BP Sleepers	7.47	7.66	4.81	66.56	68.73	118.28
1	Shorter-Sleep, High-Stress Group	6.35	6.00	7.17	40.38	73.83	128.53
2	High-Activity but High-Risk Group	6.07	6.00	8.00	90.00	75.00	140.00
3	Long-Sleep, High-Quality but Hypertensive Group	8.09	9.00	3.06	75.57	69.26	139.94
4	Moderate-Health, Elevated-BP Group	7.23	7.67	4.74	66.40	67.83	131.51
5	Long-Sleep, Low-Stress, Low-Activity Group	8.43	9.00	3.00	30.00	65.00	125.00

Table 1. Centroid-grounded names for the six K-Means clusters. Values are cluster means.

3.2 Cluster-Profile Interpretation

Six cluster profiles indicate that sleep health segmentation was not made based on sleep duration only. One of the major discriminators between clusters was the set of cardiovascular parameters, stress, and activity indicators. For instance, Cluster 1 is a profile of short sleep duration, high stress, high heart rate, and high blood pressure level. Cluster 2 plays a crucial role, as it shows high physical activity and steps per day together with high stress levels, high heart rate, and blood pressure. On the other hand, Cluster 3 and Cluster 5 are both characterized by long sleep duration and good sleep quality but differ greatly in terms of activity and blood pressure levels.

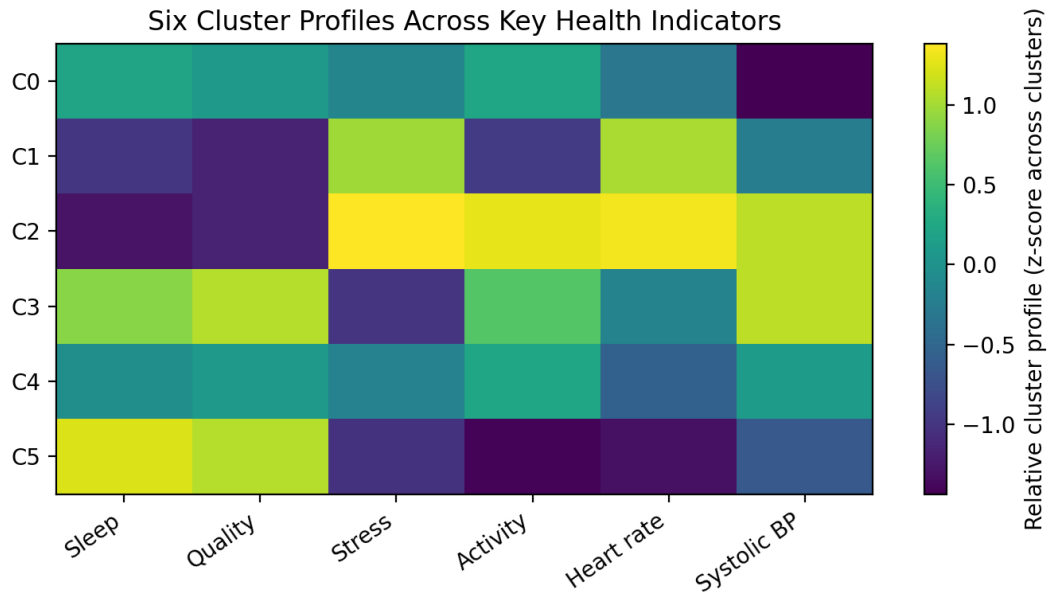


Figure 2. Relative cluster profiles across key health indicators.

3.3 PCA Visualization

The data was projected down to two principal components using PCA, which captured 69.9% of the total variation. This is quite a large percentage considering that the data was projected onto just two dimensions. Hence, the PCA chart serves the purpose of visualizing the data effectively. The groups do not collapse into one point; instead, they reside in different regions of the projection plane.

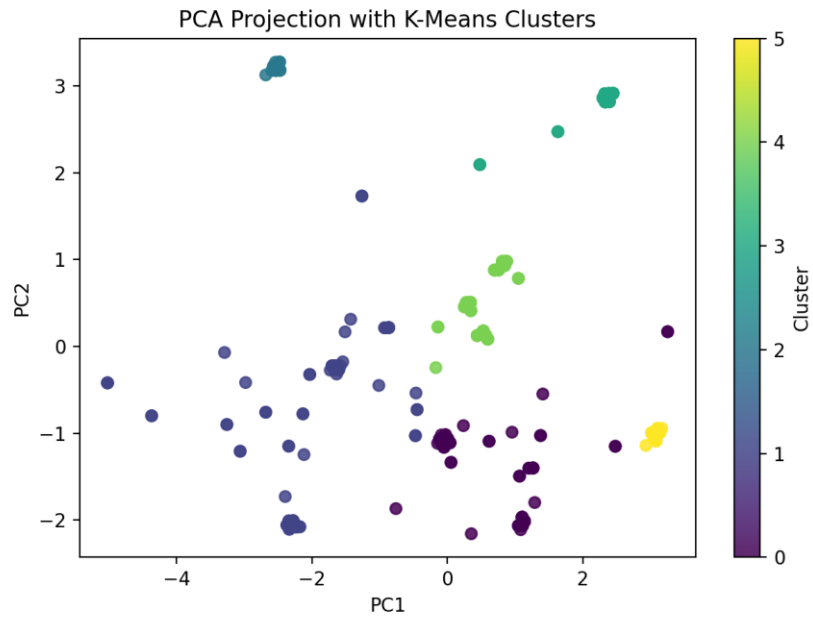


Figure 3. PCA projection of K-Means clusters. PC1 and PC2 together explain 69.9% of numeric variance.

3.4 Features That Distinguish Groups

Another Decision Tree model was used as an attempt to elucidate the K-Means cluster labels by employing the same numeric variables. However, this approach should not be considered as an external verification of the analysis, since the purpose of the tree is to learn labels generated by other related features. Rather, this is a way to enhance understanding. The four most predictive variables turned out to be systolic blood pressure, heart rate, stress level, and physical activity level. It is crucial for the analysis as a whole.

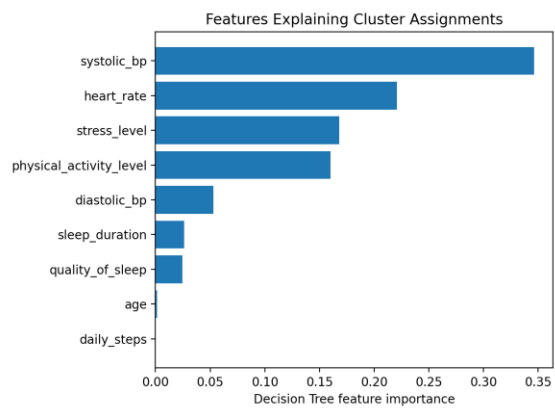


Figure 4. Decision Tree feature importances used to explain K-Means clusters.

3.5 Anomaly Detection

The Local Outlier Factor has detected 18 outliers, approximately 5% of the total data set. It would be inappropriate to regard the profiles as erroneous or ill. Local Outlier Factor detects those entries which are different compared to their neighboring entries. In this particular situation, outliers can include unusual situations such as individuals having high levels of activities but low-quality sleep and individuals who have contradictory cardiovascular and stress data compared to their neighbors.

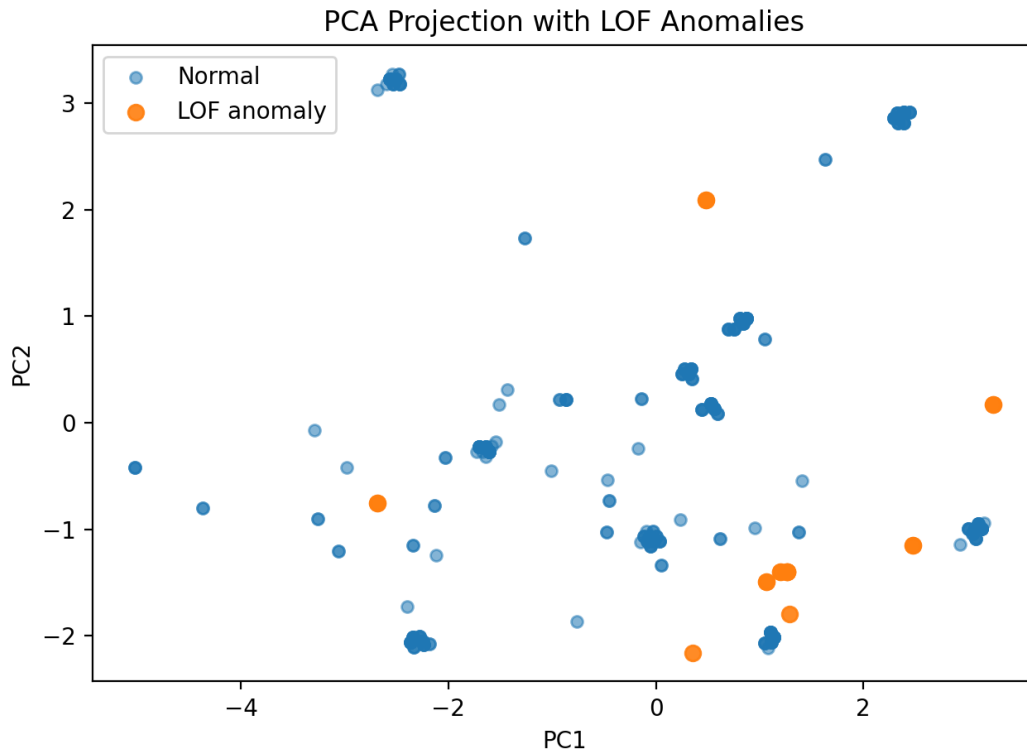


Figure 5. PCA projection with LOF anomalies highlighted.

4. Interpretation and Discovery Questions Revisited

This final interpretation links up the statistics with the initial discovery questions. In summary, the answers to all three questions have been achieved in this project, albeit with certain caveats because the study involves observational analysis.

Question	Answer	Evidence	Caveat
DQ1: Natural segments?	Fully answered: six lifestyle-health segments were identified.	K-Means $k=6$ silhouette = 0.537; Agglomerative silhouette = 0.540; PCA separation.	Other k values or features could produce different segmentations.
DQ2: Distinguishing features?	Fully answered for this feature set: blood pressure, heart rate, stress, and activity distinguish clusters most strongly.	Decision Tree importances and cluster profiles.	Tree explains cluster labels; it does not independently prove medical importance.
DQ3: Unusual profiles?	Partially answered: 18 unusual profiles were flagged for further review.	LOF with contamination = 0.05 identified 18 anomalies.	LOF flags statistical unusualness, not errors or diagnoses.

Table 2. Discovery questions revisited with answers, evidence, and caveats.

In terms of domain-specific terminologies, the key takeaway from the data set is that sleep wellness within it is multi-dimensional. Not only do the various aspects have a relationship with each other but also with sleep duration in defining the clusters within this data set. For this reason, the findings can be considered more helpful for exploration than ranking individuals' wellness statuses.

5. Critical Assessment

5.1 Validity and Robustness

The findings exhibit a moderate level of validity when applied to the current data set. This is based on several pieces of evidence, including the consistency between the K-Means and Agglomerative Clustering approaches, both of which adopt different clustering principles yet achieved relatively similar silhouette scores among six clusters. It is also worth noting that the PCA method was able to retain 69.9% of the variability using two dimensions with discernible pattern formation. Despite being promising, there are still limitations in the validation process. For example, this study did not conduct a stability test through multiple runs.

5.2 Limitations and Generalizability

- Limitations with data: The data set contains only 374 entries, which might not fully cover all age groups, jobs, medical history, or geography.
- Limitations with method used: K-Means algorithm requires compact and centroid clusters to form in the standardized numeric space. Such an approach simplifies the analysis but is rather limiting in terms of understanding.
- Limitations with preprocessing: Both standardization and feature selection might impact the outcome of the study for the distance-based analysis.
- Limitations in terms of scope: Correlations do not establish causality. The findings can neither confirm that stress causes bad sleep quality nor that high BP leads to belonging to one particular cluster.
- Limitations in terms of generalizability: The results of the work should be considered in the context of the data set only.

5.3 Ethical Considerations

Sleep and health data may be sensitive information. Regardless of the fact that this project is based on an open source dataset, proper analysis is essential. The findings cannot be employed for defining people's health status, deciding their eligibility for work positions, setting insurance risks, or giving patients recommendations about treatment. There is another issue regarding the use of occupation type, BMI group, and sleep disorders, which can be misapplied in cases of individual-level interpretations, which lack professional knowledge and social consideration. Thus, the most harmless approach to the utilization of the data is exploratory analysis, which helps reveal connections leading to further research or consultations.

6. Conclusion and Future Work

This project evolved the M3 model from its initial state to become the final data mining solution. The key finding is that the Sleep Health and Lifestyle dataset can be partitioned into six distinct groups based on their lifestyle-health characteristics. The key discriminators included systolic blood pressure, heart rate, stress level, and physical activity level, indicating that sleep-health trends are better described by wider lifestyle-health factors instead of sleep itself. PCA proved helpful for visualizing the clustering outcome, while LOF revealed 18 unique instances that need further investigation.

The most significant takeaway from this project is that pattern discovery demands much more effort than applying algorithms. This includes picking the right input data, checking outcomes through different approaches, making sense of results within an industry setting, and admitting shortcomings openly. Given more time, I would optimize the cluster labels based on additional comparisons between centroids, conduct robustness analysis using randomized seeds and features, contrast the discovered patterns against a large dataset, and analyze the 18 anomalies separately..

Repository and Reproducibility

GitHub repository: <https://github.com/kof-coder/cs4412-project.git>

For a portfolio-ready repository, the project must contain a well-written README, folders for data, notebooks, figures, and reports, the final notebook with all its output saved, a final report in PDF form, and a requirements.txt file containing necessary libraries like pandas, numpy, matplotlib, scikit-learn, and python-docx (for generating reports). The final notebook should also be executed in a fresh kernel to verify results and their visualization..

References

1. Sleep Health and Lifestyle Dataset, Kaggle. Dataset used for final implementation.
2. M. J. Zaki and W. Meira Jr., Data Mining and Machine Learning: Fundamental Concepts and Algorithms, Cambridge University Press.
3. Scikit-learn documentation for K-Means, Agglomerative Clustering, DBSCAN, PCA, DecisionTreeClassifier, GaussianNB, and LocalOutlierFactor.